



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 61



**BOWA Multi-Purpose Shredder
Model: GR-1300**

BOWA GR-1300 MULTI-PURPOSE SHREDDER

Test Report No.	:	2013 - 61
Date of Test	:	March 7, 2013
Test Valid Until	:	March 7, 2018
Brand	:	BOWA
Model	:	GR-1300
Type	:	Open cylinder, horizontal single shaft
Manufacturer	:	KAINITZ CORPORATION 452 Gen. Bautista St., San Juan City
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The BOWA GR-1300 Multi-Purpose Shredder was tested using split “buko” husk with shell with an average Moisture Content of 81.9%.

The BOWA GR-1300 Multi-Purpose Shredder had an average Input Capacity of 1886 kg/h of “Buko” husk with shell at blade shaft speed of 3338 rpm

The Fuel Consumption Rate of the 9.70 kW BOWA BG390F gasoline engine was 2.53 L/h.

Two persons were needed to operate the machine.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted at the Agricultural Machinery Testing and Evaluation Center (AMTEC) in U. P. Los Baños, College, Laguna on March 7, 2013. The test was carried out to assess and verify the following::

1. Machine specifications
2. Performance
 - a. Input capacity
 - b. Speed of components
 - c. Noise emitted by the machine
 - d. Fuel Consumption
3. Labor requirement

OPERATOR'S MANUAL

No operator's manual or brochure of the machine was submitted.

METHOD OF TEST

The shredder was tested following the “PAES 245-2010, Agricultural Machinery – Biomass Shredders – Methods of Test”.

DESCRIPTION OF THE MACHINE

The BOWA GR-1300 as shown in **Figure 1** was an open cylinder, horizontal single shaft shredder/chipper. It was powered by a 9.70 kW BOWA BG390F gasoline engine which drove directly the machine's blade system through a belt-pulley transmission system (**Fig. 2**). The machine consisted of the hopper and the shredding/chipping chamber.

The hopper (**Fig. 3**) was where the materials to be shred are fed. Material to be shred was fed into the hopper to a rotating cylinder with blades and ejected to an outlet chute (**Fig. 4**) located at the bottom of the shredding cylinder in the opposite side of the hopper.

The shredding/chipping chamber (**Fig. 5**) consisted of a horizontal, single, open type cylinder, a chipping plate and housing. The shredding cylinder had a width of 295 mm and a diameter of 328 mm and had four longitudinal anchor bars. Each anchor bar was provided with eight and seven (alternate) swinging hammers (**Fig. 5a**) made of hot rolled steel #45 and had dimensions (L x W x T) 85 mm x 38 mm x 5 mm. The hammers were alternately arranged with respect to the hammers in the next anchor bar. At one end of the cylinder was a chipper plate with two blades (**Fig. 5b**). Materials (i.e. woods) to be chipped were fed to an 84 mm diameter opening (**Fig. 6**) which was provided at the side of the machine. The shredding/chipping chamber was provided with a replaceable screen which served as its lower concave (**Fig. 5c**).

It was also provided with four pieces of rubber wheels (**Fig. 7**) which were used during transport.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. BOWA Multi-Purpose Shredder
Model: GR-1300

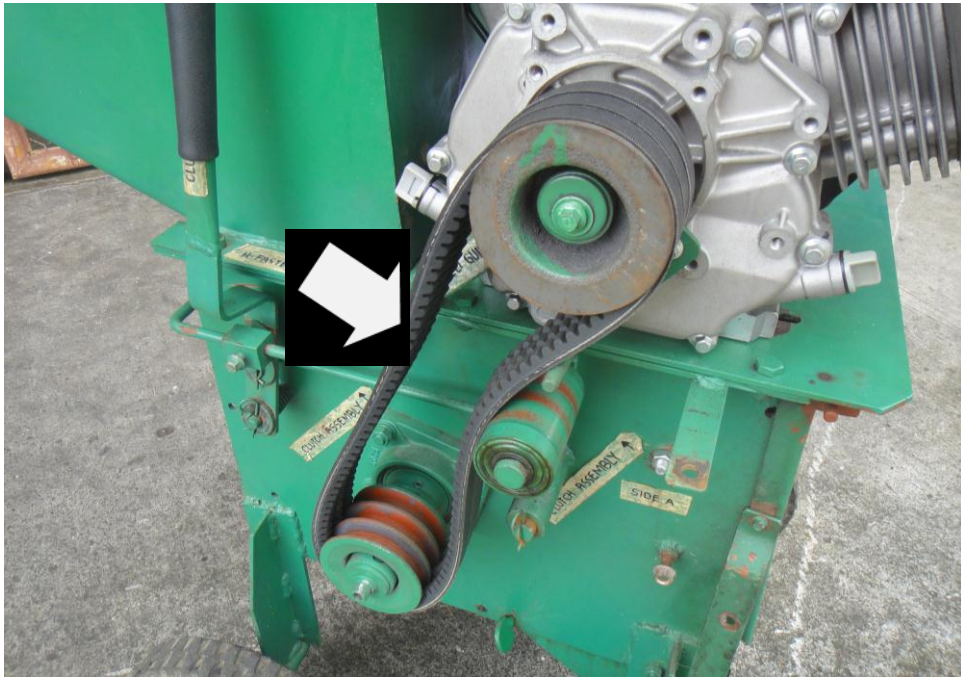


Figure 2. Belt-pulley transmission system



Figure 3. Hopper



Figure 4. Outlet chute

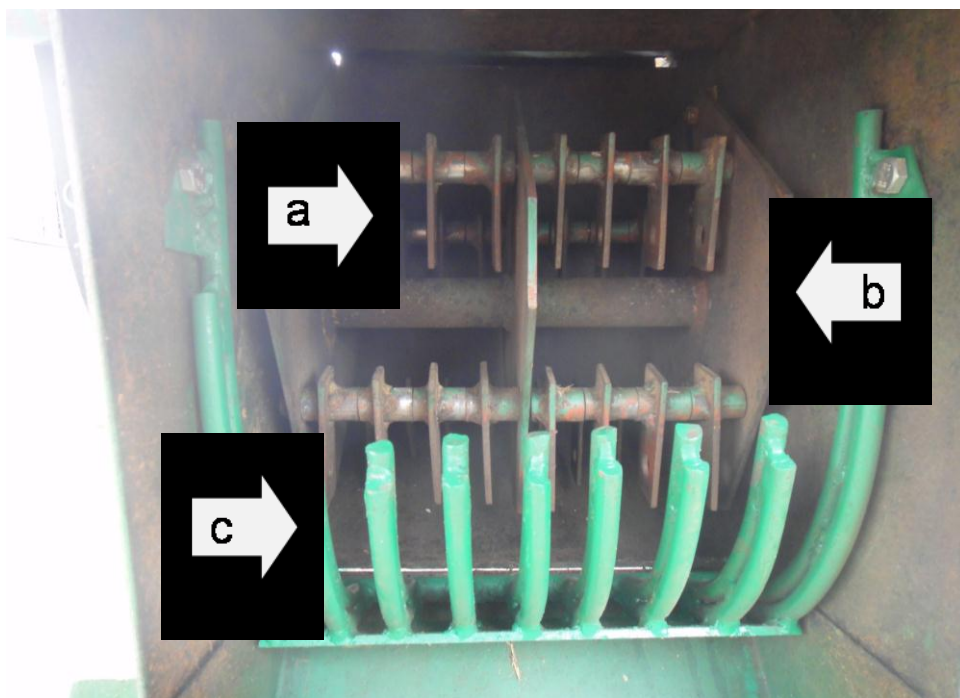


Figure 5. Shredding/Chipping Chamber
a) Swinging hammer
b) Chipper plate with blades
c) Lower concave

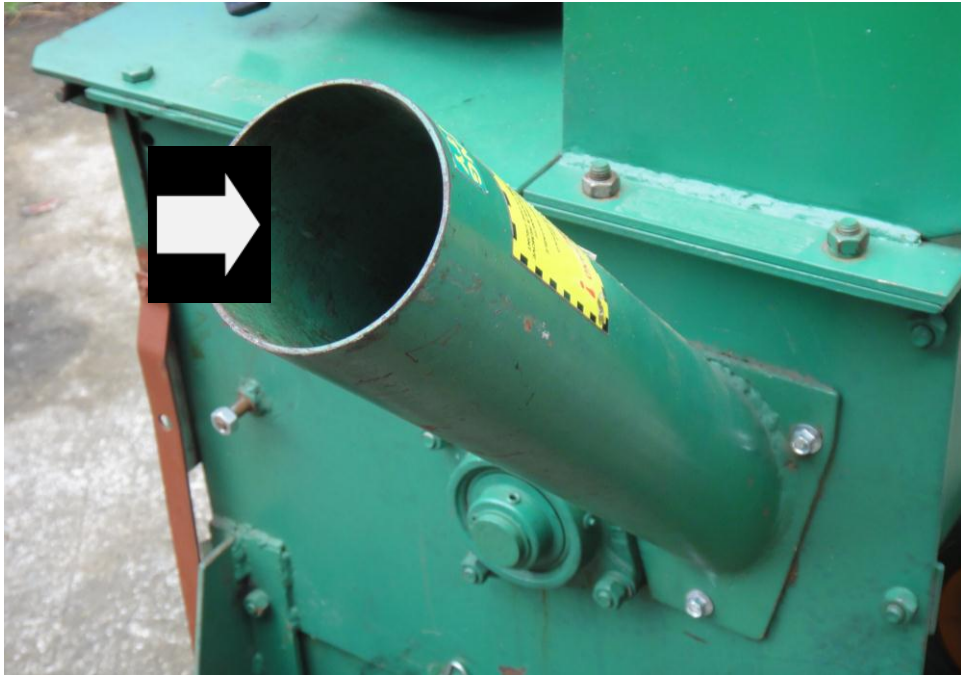


Figure 6. Chipping inlet opening

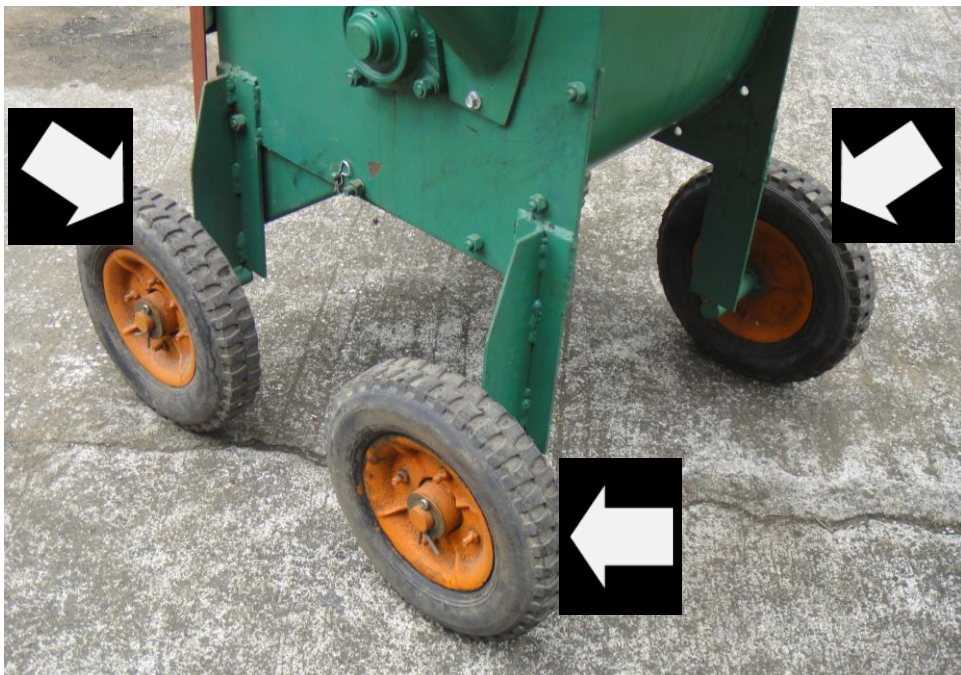


Figure 7. Rubber wheels

SPECIFICATIONS

Table 1. Detailed specifications of the BOWA GR-1300 Multi-Purpose Shredder

A.	Brand	:	BOWA
B.	Model	:	GR-1300
C.	Serial No.	:	-
D.	Manufacturer	:	KAINITZ CORPORATION 452 Gen. Bautista St., San Juan City, Metro Manila
E.	Overall Dimensions and Weight		
1.	Length (mm)	:	1365
2.	Width (mm)	:	830
3.	Height (mm)	:	1240
4.	Weight (kg) without engine	:	134.0
F.	Hopper		
1.	Bottom opening, L x W (mm)	:	335 x 80
2.	Material	:	M. S. Plate (2.5 mm thickness)
3.	Height from the ground	:	1125
G.	Shredding Assembly		
1.	Type	:	Open cylinder

Table 1. Detailed specifications of the BOWA GR-1300 Multi-Purpose Shredder (cont'n.)

2.	Size, W x D (mm)	:	295 x 328
3.	No. of anchor bars	:	4
4.	Hammer		
a)	Type	:	Swinging
b)	No. per anchor bar	:	8, 7, 8, 7
c)	Arrangement	:	Alternate with respect to the hammers in the next anchor bar
d)	Size, L x W x T (mm)	:	85 x 38 x 5
e)	Means of attachment	:	Hanging through hole
f)	Material	:	Hot Rolled Steel #45
H.	Chipper Plate		
1.	Material	:	M. S. Plate
2.	No. of blades	:	2
3.	Size of blades, L x W x T (mm)	:	105 x 40 x 5
4.	Chipper opening, D (mm)	:	84
I.	Lower Concave		
1.	Type	:	Bar Screen
2.	Spacing (mm)	:	22
3.	Material	:	Mild Steel (13 mm diameter)

Table 1. Detailed specifications of the BOWA GR-1300 Multi-Purpose Shredder (cont'n.)

J.	Cylinder cover material	:	M. S. Plate (5.5 mm thickness)
K.	Output chute		
	1. Size of opening, H x W (mm)	:	330 x 265
	2. Height from the ground (mm)	:	165
L.	Belt-Pulley Transmission System		
	1. Engine pulley (mm)	:	102 x 3B x 35
	2. Shredding shaft pulley (mm)	:	89 x 3B x 25
	3. Belt	:	B-54
M.	Transport Wheels		
	1. No. of pieces	:	4
	2. Size, D x T, mm	:	295 x 66
N.	Standard Accessories		
	1. Wet Screen	:	Bar screen with 32 mm Spacing
	2. Dry Screen	:	Perforated screen with 38 mm perforations
O.	Optional Accessories	:	Perforated screen (from 2 mm to 13 mm perforations)
P.	Engine (Prime mover)		
	1. Brand	:	BOWA
	2. Model	:	BG390F

Table 1. Detailed specifications of the BOWA GR-1300 Multi-Purpose Shredder
(cont'n.)

3.	Make	:	China
4.	Type	:	Four-stroke, OHV, single cylinder gasoline engine
5.	Rated Maximum Power (kW)	:	9.70 @ 3600 rpm
6.	Cooling system	:	Air-cooled
7.	Starting system	:	Rope recoil
8.	Weight (kg)	:	33.0

RESULTS AND DISCUSSIONS

Results of performance tests are shown in **Table 3**.

Input Capacity

Test results showed that the machine had an Input Capacity of 1886 kg/h of split “buko” husk with shell at blade shaft speed of 3338 rpm.

Fuel Consumption

The 9.70 kW BOWA BG390F gasoline engine had a Fuel Consumption Rate of 2.53L/h.

Labor Requirement

Two persons were needed to operate the machine (**Fig. 8**).

Observations:

1. The shredder was provided with belt guard (**Fig. 9**).
2. Split “buko” husks with shell (**Fig. 10**) were used in the test. Totally shredded coconut husks (**Fig. 11**) were produced.
3. The shredder was provided with a handle which was used for transport (**Fig. 12**).
4. The shredder was provided with safety stickers (**Fig. 13**).



Figure 8. Operators of the machine



Figure 9. Belt guard



Figure 10. “Buko” husks with shell



Figure 11. Shredded coconut husks



Figure 12. Handles used for transport



Figure 13. One of the safety stickers attached to the machine

PERFORMANCE TEST

Machine Tested : BOWA Multi Purpose Shredder
Model: GR-1300

Date of Test : March 7, 2013

Location of Test : AMTEC
U. P. Los Baños, College, Laguna

Test Material Used : Spllit “Buko” husk with shell

Table 2. Conditions of Test Materials

ITEMS	Data
1. Moisture Content, MC _{w.b.} (%)	81.9
2. Dimensions, L x D (mm)	182 x 135

Table 3. Result of Performance Test

ITEMS	Data
1. Speed of Components	
a) Engine (rpm)	
Without load	3139
With load	3012
b) Blade Assembly Shaft	
Without load	3530
With load	3338
2. Noise Level, dB(A)	
Without load	95.6
With load	99.0
3. Prime mover's Fuel Consumption (BOWA BG390F Gasoline Engine), L/h	2.53
4. Input Capacity, kg/h	1886

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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